Nearest Neighbor Reasoning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Nearest-Neighbor Reasoning

> Prepare Nearest Neighbor Reasoning: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Compute absolute **distance**s and rank **training neighbour**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/22, ai-4/12. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Define a question and target population

Reference example, not the only acceptable wording: Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

Overfitting Intuition

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

Concept to explain

A new item can be predicted from nearby labeled examples.

Term	Meaning
neighbor	An example close to another under a specified distance measure.
k	In k-nearest neighbours, the number of nearby examples considered in the prediction.
vote	A contribution to a choice, such as one neighbour's label in a classification decision.
tie rule	A predefined method for handling candidates with equal scores or conflicting nearest labels.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Use a single numeric feature to make **distance** arithmetic visible, then distinguish **neighbour** selection from voting. Store only **training** rows.

Alphabetical IDs settle equal **distance**s for this classroom method; explain that it is a convention and other implementations can use different tie behavior.

Readiness answer

It stores a root question and leaf **prediction**s learned from **training**. A new example follows the appropriate branch to a stored leaf **prediction**.

Printed lesson and workbook

Number line from 0 to 14 cm and five labelled **training** markers

Optional Python 3 editor with no packages required

Blank table for **distance**s ordered IDs **vote**s and **prediction**s

End goal for the pictorial case

By the end, I can order **training neighbour**s, apply k and keep query labels out of **prediction**.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Nearest Neighbor Reasoning using the supplied case, labelled mechanism and

> reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

> 1. New book's 3 nearest labels: adventure, adventure, mystery.

> 2. With k=3, adventure wins 2 **vote**s to 1.

> 3. With k=1, the closest single book controls the result.

> 4. Choose k and **tie rule** before viewing the desired answer.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

The teaching case

Find nearest **neighbour**s using differing-trait **distance** and k=1.

Exact explanation to model

Why the result follows

A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Explain the added worked example

Explain the added worked example · 1

A nearest-**neighbour** classifier predicts using nearby **training** examples rather than following a fitted tree.

This week we store labelled examples, measure how different their feature values are from a new query, and let selected **neighbour**s **vote**.

Nearby means close under our chosen feature and **distance**, not necessarily similar in every real-world way.

Explain the added worked example · 2

Our new fictional dataset has a **distance** reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity.

IDs identify rows and break **distance** ties; they are not numeric features. These invented readings are not a robot safety system.

Explain the added worked example · 3

For this one-dimensional feature, **distance** is abs(**training** reading minus query reading). abs removes the sign. Set k=3 and give each selected **neighbour** one **vote**.

Order equal **distance**s by ID alphabetically, declared before scoring. In this binary, three-**vote** exercise the label **vote** cannot tie, but the **distance**s can.

For query 3, **distance**s to A through E are 1,1,5,7,9 cm. sorted orders tuples by **distance** first, then the unique ID. ranked[:3] takes the first three rows.

The **output** is ['A', 'B', 'C'] then clear: one blocked **vote** and two clear **vote**s.

If k were 1 under the same **distance tie rule**, A would win and predict blocked. Changing k can change a **prediction**.

We declared k=3 for this exercise; do not choose k after seeing the desired answer and pretend the method was unchanged.

Explain the added worked example · 4

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three **training** rows. It does not validate those conditions.

A larger k than the stored row count would not produce the promised number of **vote**s. Keep unresolved review labels under a separate documented policy.

Key terms

Term	Meaning
Distance	A numerical measure of difference between selected feature values
Neighbour	A stored training example close under the chosen distance
k	The number of neighbours selected for a prediction
Vote	A contribution of a selected neighbour’s label to the predicted class

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Three selected neighbours for query 3

STORED EXAMPLE	FEATURE DISTANCE	ONE LABEL VOTE
A at 2 cm	abs 2 minus 3 is 1	blocked
B at 4 cm	abs 4 minus 3 is 1	clear
C at 8 cm	abs 8 minus 3 is 5	clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Model the reasoning aloud · 5

Store the **training** examples and the method settings. At **prediction** time, compare the query only with stored **training** examples. Do not add the query's reference label or other validation labels to the **neighbour** pool.

No cutoff or tree is fitted here, but retaining **training** examples is still part of the learning method.

> **Why do A and B tie for distance to query 3? Which wins for k=1 under our rule, and why does k=3 predict differently? Does the ID rule establish which label is true?**

A **vote** share of two out of three describes these **neighbour**s. It is not a verified two-thirds probability that the new scene is clear. A **prediction** can be wrong even when the nearest examples agree.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Nearest Neighbor Reasoning

A fictional one-feature classifier predicts compact or extended book format.

The facts we will use

Supplied facts

Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

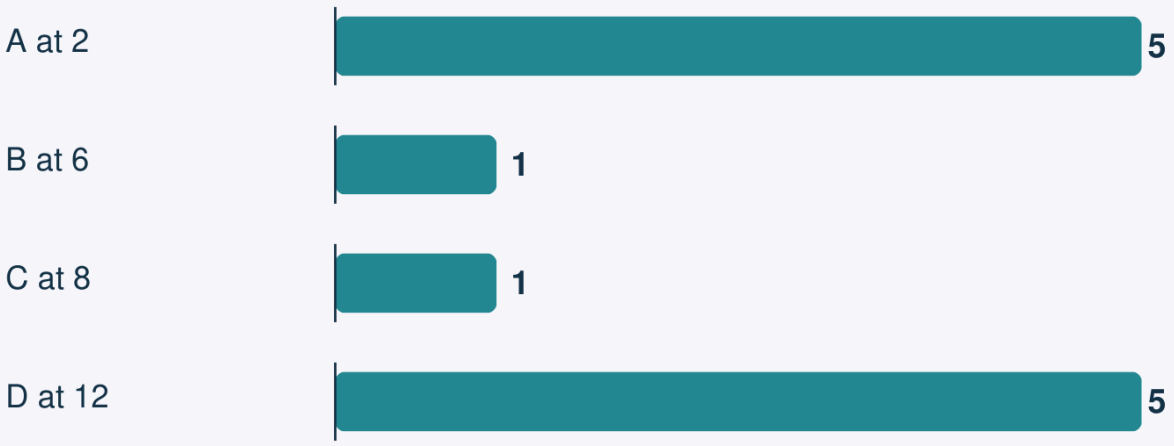
Supplied facts

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

AI ACADEMY / AI-2/22

Distances from query 7

absolute distance units



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/22

Nearest Neighbor Reasoning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

From 7: A=5, B=1, C=1, D=5.



2. Order neighbours

Distance then ID gives B,C,A,D.



3. k=1 prediction

First neighbour B is extended: predict extended.



4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Measure distances	From 7: A=5, B=1, C=1, D=5.
2. Order neighbours	Distance then ID gives B,C,A,D.
3. k=1 prediction	First neighbour B is extended: predict extended.
4. k=3 prediction	B,C,A vote extended,compact,compact: predict compact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

What this example does not establish

Keep the stated limit

Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

Change one thing in the pictured case

Change one thing

Change: Move query from 7 to 9 while keeping **training** fixed.

Predicted result

Prediction: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

Why the result follows

Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Nearest Neighbor Reasoning, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

2. State the decision to explain

Write the question to explain: Find nearest **neighbour**s using differing-trait **distance** and k=1.

3. Reveal the worked reasoning

Reveal after **prediction**s: A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Hold back the independent task

Keep for an independent attempt: **Training** items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute **distance** to choose its nearest **neighbour**.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: abs(2-3) and abs(4-3) both equal 1. For k=1 alphabetical tie order chooses A, blocked. k=3 adds B and C, whose clear **vote**s win 2 to 1. ID order is a deterministic convention, not **evidence** of the query's true label.

Exit: using labelled validation examples in the **neighbour** pool leaks evaluation answers and may let a query select itself. Restore the original **training**-only pool and score every validation query without adding its label.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve tree storage and prediction. Introduce prediction from stored examples with a declared distance.
7 to 18 minutes	Place A through E on the number line. Calculate absolute distances to query 3 and complete W1.

Stage	Teacher action
18 to 32 minutes	Trace tuple sorting and list slicing. Complete W2; distinguish tied distances from a tied label vote.
32 to 45 minutes	Read the diagram and answer the guided question. Explain why ID order is not evidence of truth.
Session 2 start to 8 minutes	Retrieve the training-only neighbour pool. Ask why validation labels must remain outside it.
8 to 20 minutes	Compute the three validation predictions and complete W3. Compare the fixed training-majority baseline.
20 to 35 minutes	Collect W4 independently before support. Require ordered IDs, both k predictions and the full validation score.
35 to 45 minutes	Use W5 and the exit question. Offer the fresh retry after feedback and record support.

Reduce help deliberately

Completion practice

Completion task: For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Reasoning to listen for

Teacher check: For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID **tie rule** stay the same; the number of voting **neighbour**s changes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Collect explanations, not only yes/no **vote**s. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why do k=1 and k=3 disagree for the original query?

Reasoned teacher answer

The closest tied **neighbour** selected by ID is extended B, while the first three include two compact **vote**s.

Changed-case reasoning: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Nearest Neighbor Reasoning investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Measure **distance**s, circle k=3 nearest, tally **vote**s, apply a written **tie rule**, and test one point near the boundary.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A negative difference is a negative **distance**: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Odd k prevents every kind of tie: Three binary **vote**s avoid a label tie, but **neighbour**s can still tie in **distance**, including at the selection boundary.

An exact match proves the label: It matches the selected feature only. Other conditions, annotation errors or contradictory examples can remain.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Nearest Neighbor Reasoning questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Independent question

Independent question · withhold the key

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute **distance** to choose its nearest **neighbour**.

Expected independent answer

Teacher key · Expected independent answer

Distances are 4 to A and 2 to B, so nearest-**neighbour prediction** is long. This follows the supplied **distance** rule, not certainty about every future case.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The first stored example is always the nearest.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> Row-order choice makes **prediction**s change when data is reordered. Predefine review for conflicting nearest labels, or a justified stable tie-breaker with disclosed limits.
>
> Do not select a tie policy after looking at the correct answer for the test case.

Explanations for the additional workbook examples

W1 Calculate feature distances

Workbook question · W1

For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

Teacher reasoning · W1

Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the **distance** calculation.

W2 Apply neighbour and tie rules

Workbook question · W2

For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Teacher reasoning · W2

For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID **tie rule** stay the same; the number of voting **neighbour**s changes.

W3 Keep evaluation examples separate

Workbook question · W3

Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed

k=3 score and baseline score on the worked validation examples.

Teacher reasoning · W3

Adding validation labels uses evaluation answers to make **prediction**s and changes the intended comparison with unseen trips. The fixed k=3 method predicts clear three times and gets 1/3.

The **training**-majority clear baseline also gets 1/3.

W4 Find neighbours independently

Workbook question · W4

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

Teacher reasoning · W4

Query 6 **distance**s P,Q,R,S,T are 5,1,1,5,7. Order Q,R,P,S,T. k=1 predicts blocked from Q; k=3 uses Q,R,P and also predicts blocked. For queries 6,10,12, selected triples are Q R P, S R T, S T R.

All predict blocked, matching validation rows 1 and 3: 2/3. **Training**-majority blocked also scores 2/3.

W5 Interpret a vote and a match

Workbook question · W5

Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

Teacher reasoning · W5

The **vote** fraction describes the selected **training** labels, not a verified probability of correctness. An exact match on one feature can still differ in omitted conditions or have a wrong or conflicting label.

Separate evaluation and relevant features remain necessary.

Assess independent transfer

Independent question · withhold the key

Selected case: New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked.

For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3. Using the same **training** pool and k=3, score validation (6,blocked), (10,clear),(12,blocked).

Compare the **training**-majority baseline.

Selected case answer

Teacher key · Selected case answer

Query 6 **distance**s P,Q,R,S,T are 5,1,1,5,7. Order Q,R,P,S,T. k=1 predicts blocked from Q; k=3 uses Q,R,P and also predicts blocked. For queries 6,10,12, selected triples are Q R P, S R T, S T R.

All predict blocked, matching validation rows 1 and 3: 2/3. **Training**-majority blocked also scores 2/3.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

Connect Nearest Neighbor Reasoning to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Use **neighbor**ing liked books as **evidence**, while explaining which traits created closeness.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 23: Recommendation Scores. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use physical steps on a number line before abs. Supply an empty **distance** table without the order.

Extension: explain how renaming tied IDs can change a **prediction** under this policy while the measured features remain unchanged, and why the policy must be recorded.

Require W4 **distance**s 5,1,1,5,7, order Q,R,P,S,T, both **prediction**s blocked, and validation plus baseline scores 2/3. Fresh retry: L at 0 blocked, M at 4 clear, N at 6 clear, O at 10 blocked, P at 12 clear; query 5.

Distances are 5,1,1,5,7; order M,N,L,O,P. k=1 is clear from M; k=3 is clear from M,N,L. Ask why L wins the third-place **distance** tie over O: alphabetical ID policy. Require a **training**-only pool and no probability claim.

Week 11 covers scaling without leakage. Carry forward **training**-only storage and explicit **distance** choices.

Teacher reference: https://scikit-learn.org/stable/modules/**neighbor**s.html (nearest-**neighbour** classification and tie considerations; reviewed 2026-09-27).

Plan the delayed return

Suggested return points in this level: ai-2/23, ai-2/25, ai-2/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Nearest Neighbor Reasoning lesson; it

> is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Compute absolute **distance**s and rank **training neighbour**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Supported practice answer

Reasoned practice answer: For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID **tie rule** stay the same; the number of voting **neighbour**s changes.

Check every learner before discussion: Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Move query from 7 to 9 while keeping **training** fixed. Require a **prediction** and a reason before comparing. Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

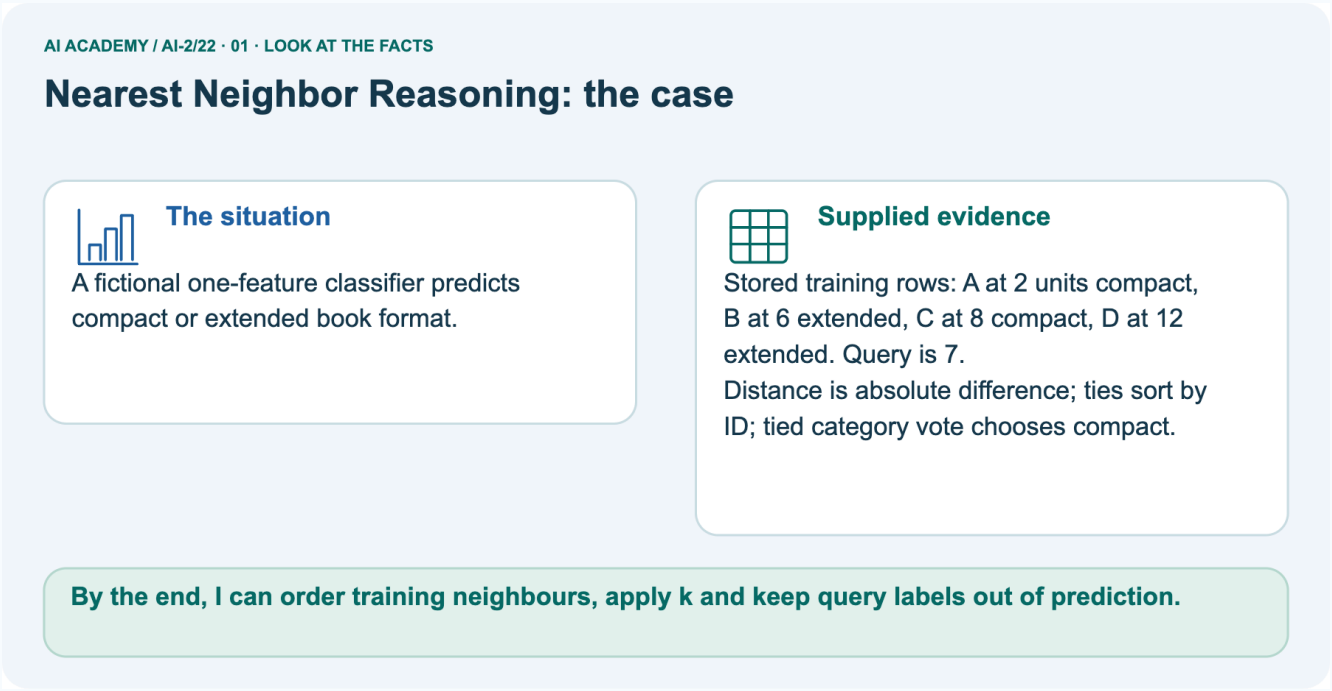
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Nearest Neighbor Reasoning graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Nearest Neighbor Reasoning: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional one-feature classifier predicts compact or extended book format. Supplied facts: Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

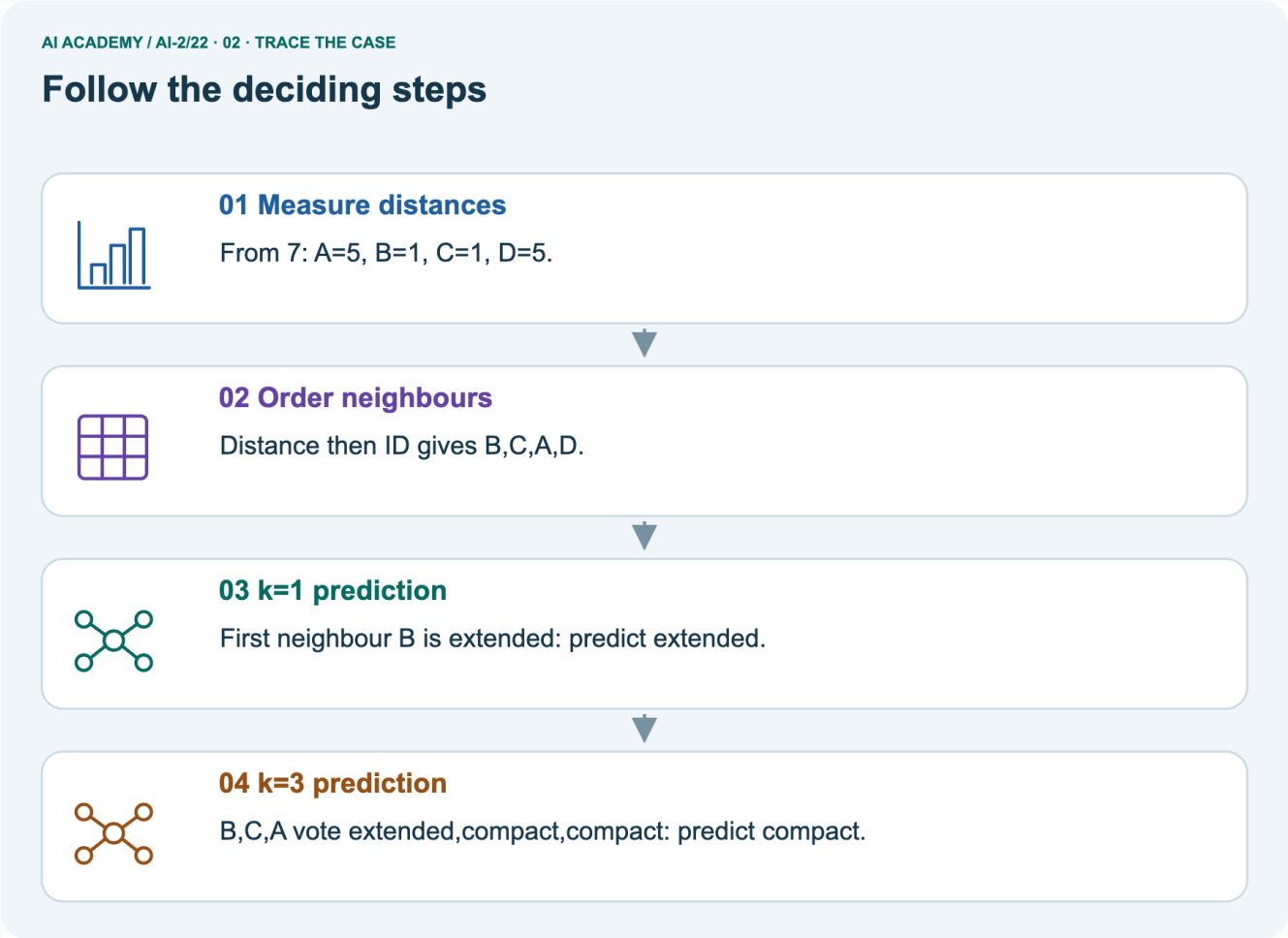
Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

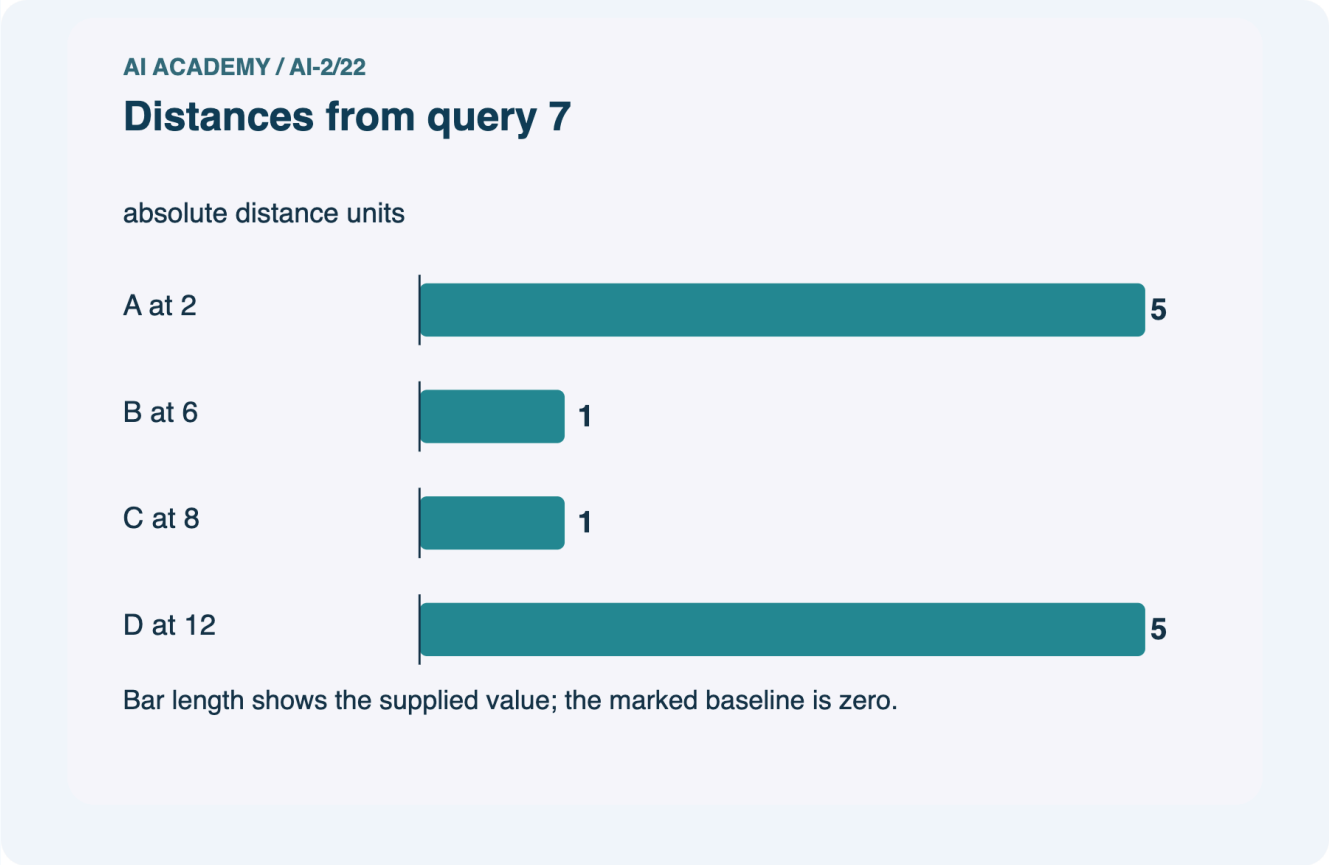
Measure **distance**s: From 7: A=5, B=1, C=1, D=5. → Order **neighbour**s: **Distance** then ID gives B,C,A,D. → k=1 **prediction**: First **neighbour** B is extended: predict extended. → k=3 **prediction**: B,C,A **vote** extended,compact,compact: predict compact.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Distances from query 7



Follow the labelled picture with the matching explanation. Open the vector diagram

Distances from query 7. Units: absolute **distance** units. A at 2: 5. B at 6: 1. C at 8: 1. D at 12: 5.

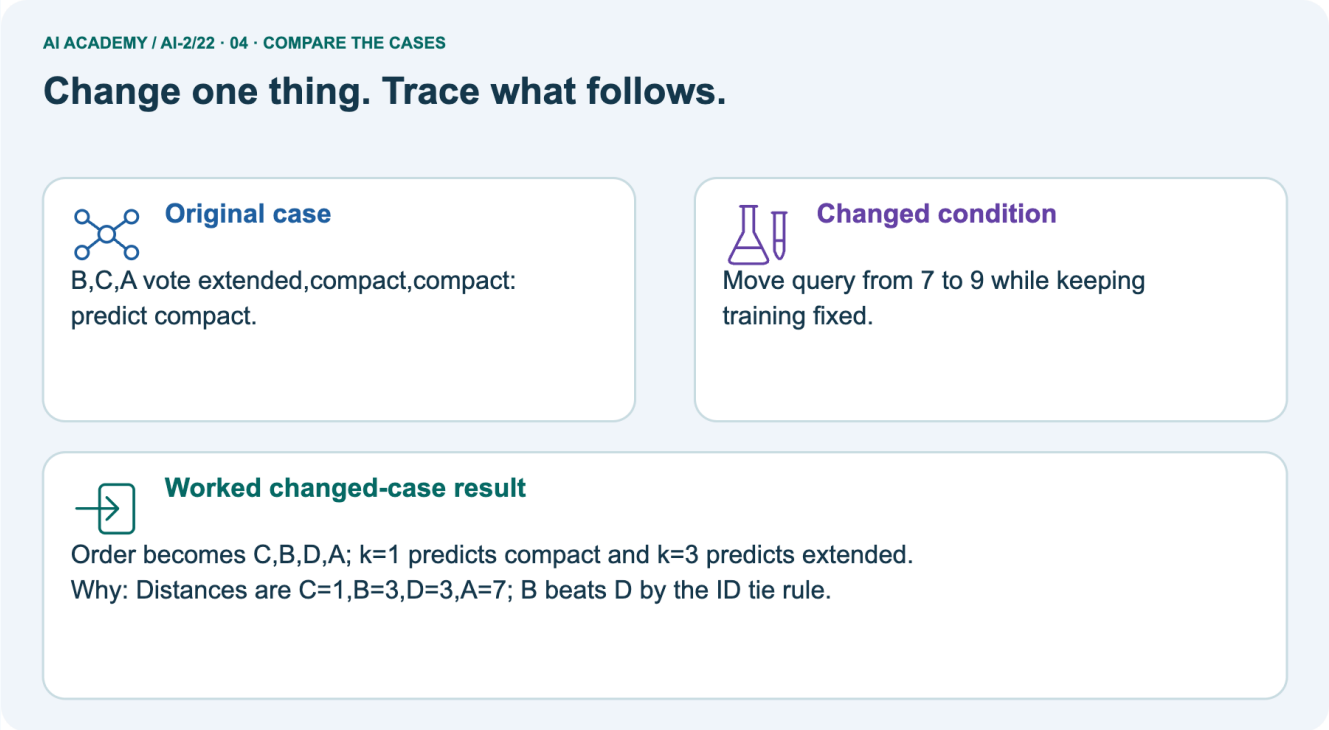
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B,C,A **vote** extended,compact,compact: predict compact. Changed condition: Move query from 7 to 9 while keeping **training** fixed.

Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

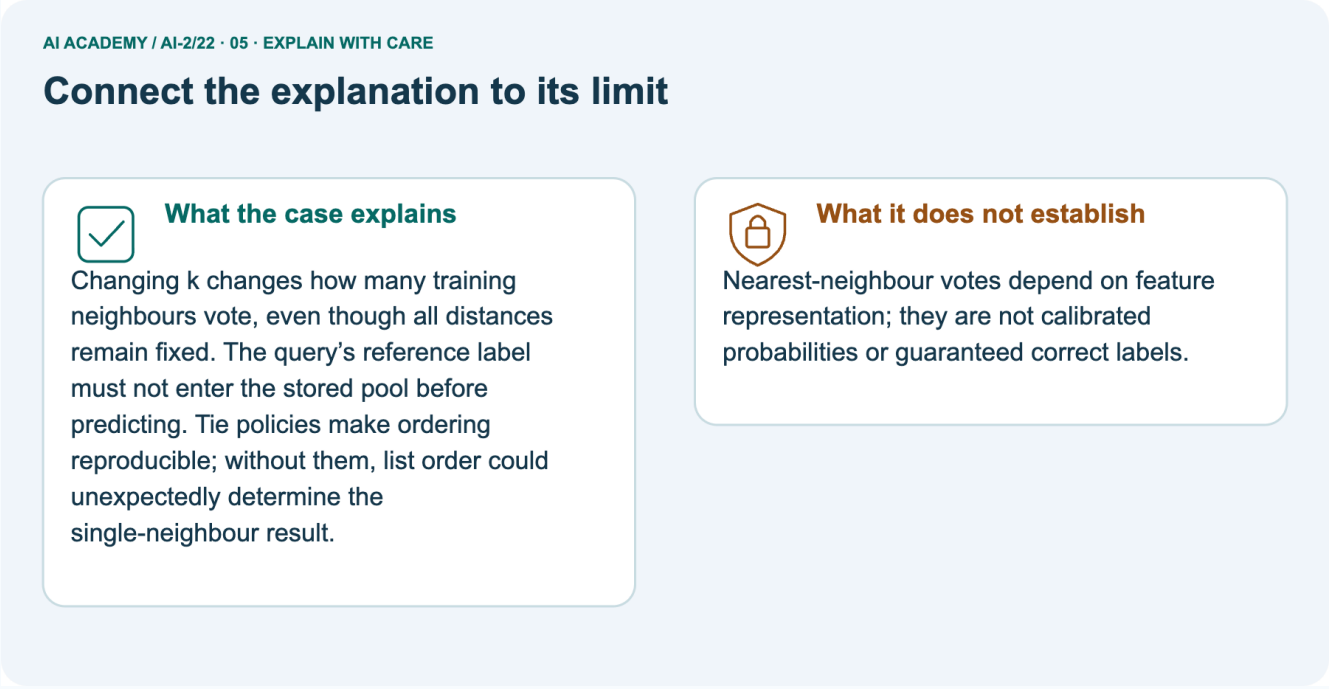
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Limit: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

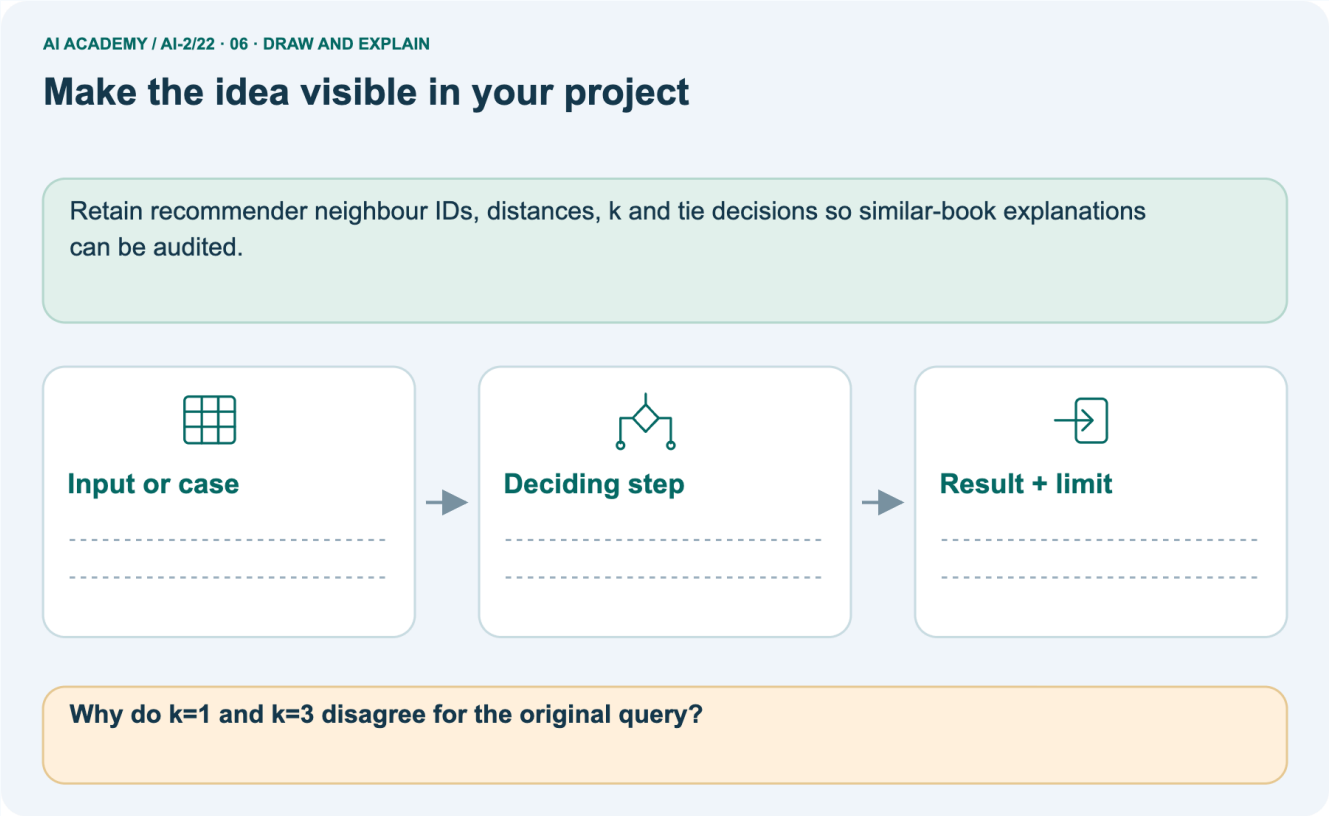
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited. Draw your **input**, deciding step, result and limit.

Supported question: Why do k=1 and k=3 disagree for the original query?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The closest tied **neighbour** selected by ID is extended B, while the first three include two compact **vote**s.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.